



Current class diagrams of the game

This document describes the classes and runtime boundaries inferred from the current source code in `include/` and `src/`. The diagrams are split by responsibility so each slice can be read without confusing ownership relations with usage relations.

Scope and how to read

- `*--` is composition: the left object owns the component's lifetime (usually a by-value member or a `std::unique_ptr`).
- `o--` is aggregation/non-owning recording; `-->` and `..>` are references, calls, or return types — they transfer no ownership.
- Cardinality labels (`1` , `0..1` , `0..*` , `2`) reflect the runtime members/collections, not the number of statically created C++ classes.
`<<RAII token>>` is a role declared or shown directly in the code.
`SaveManager` deliberately carries no Singleton label.
- Class, enum, and method names stay in English as in C++; the explanations are prose. Long templates are shortened inside class boxes to keep Mermaid stable; the precise owning type is recorded in the notes and evidence table.

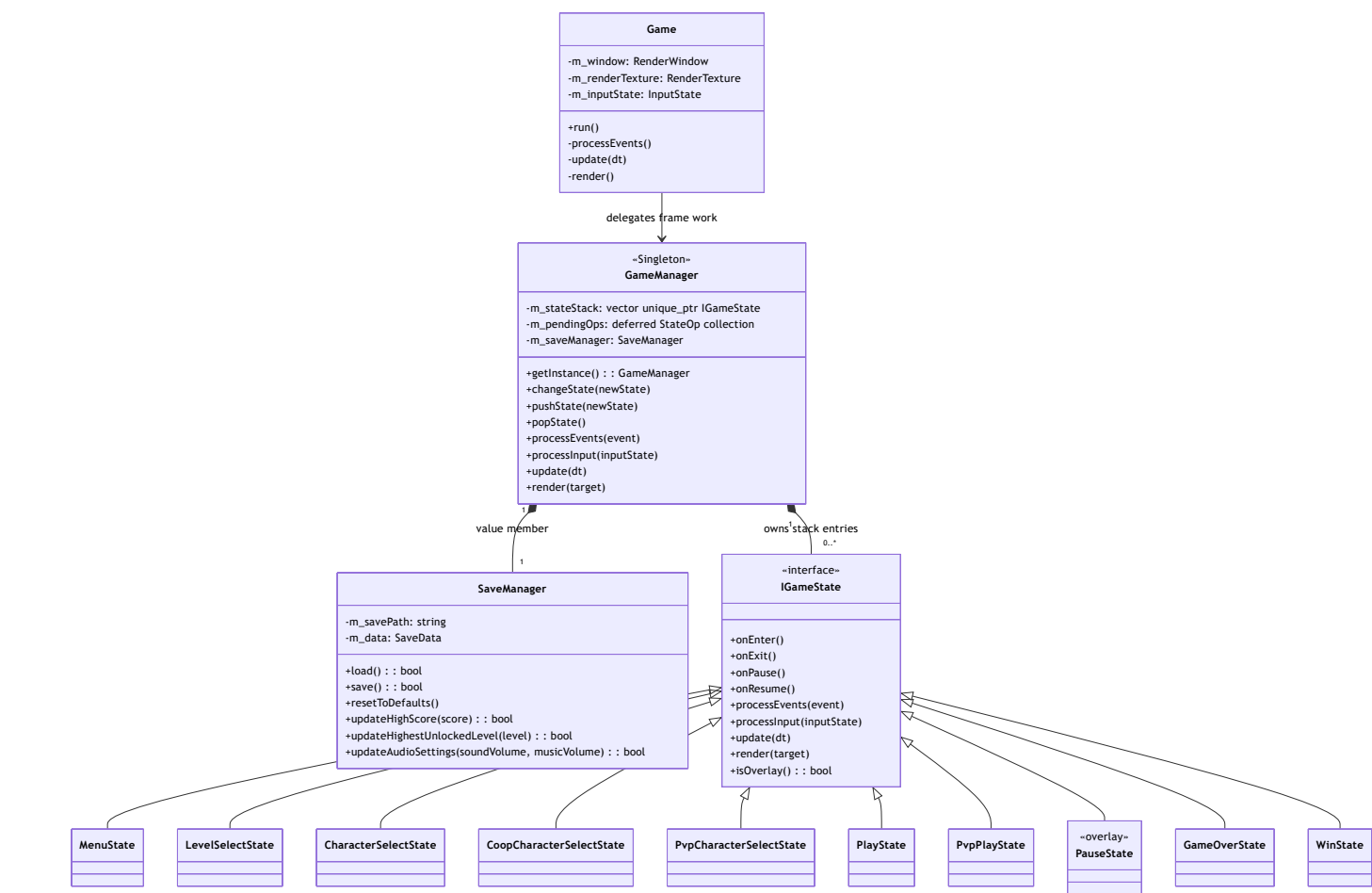
The color scheme is consistent across blocks: orange = core/owner, blue = state or physics, green = gameplay/entity, purple = pattern/event, and light red = enemy. Colors are only a visual aid; semantics are decided by the arrows and notes.

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1. Application loop and state stack

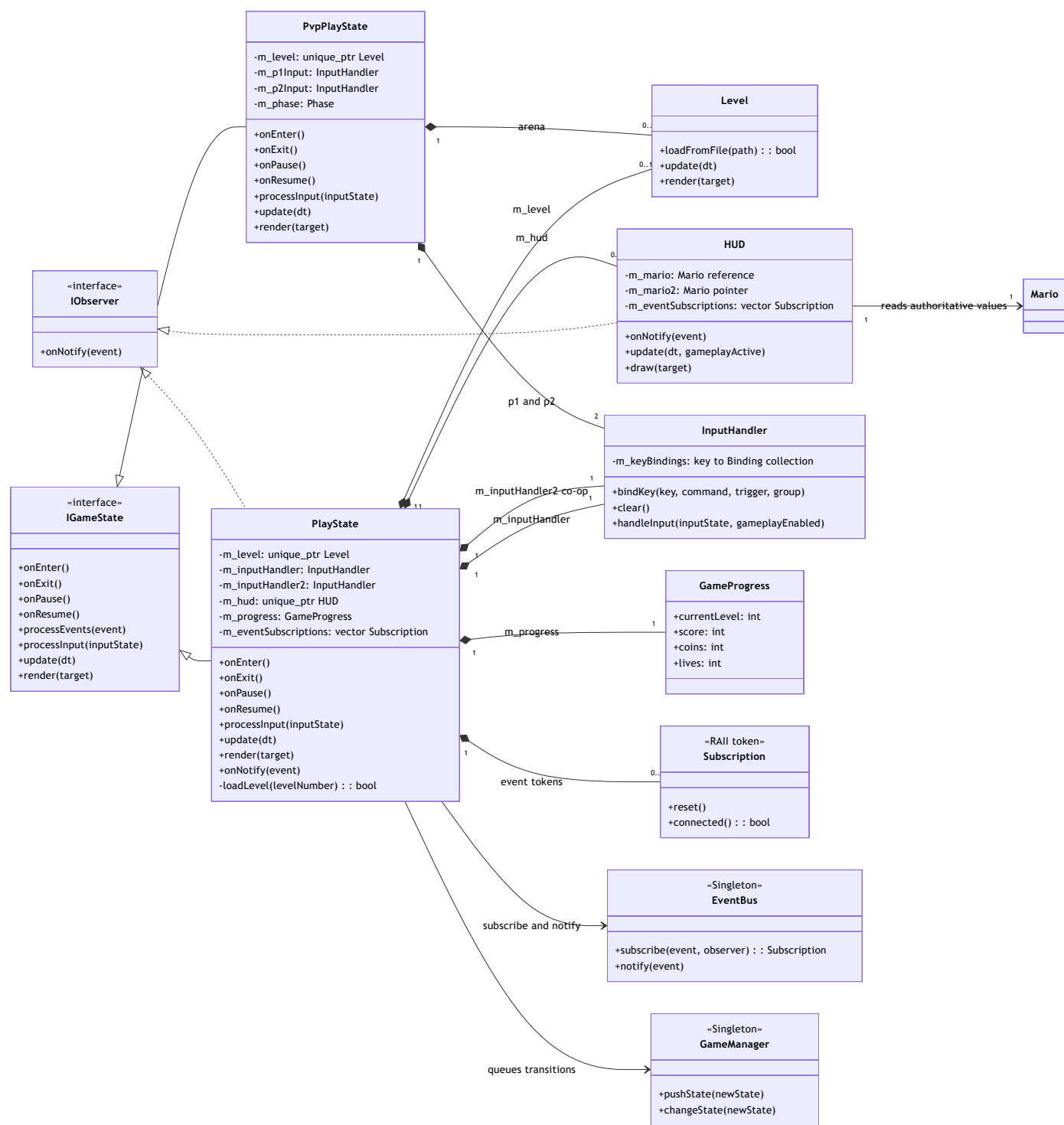
This is the path of one frame: `Game` receives events and keeps the `InputState`, then delegates input, update, and render to `GameManager`.
`GameManager` is a Singleton coordinating a stack of `unique_ptr<IGameState>`;
`SaveManager` is merely a by-value member of the manager, not an independent Singleton.



`GameManager::changeState`, `pushState`, and `popState` only enqueue a `PendingOp`; only at the end of `GameManager::update()` does `processPendingOps()` run. `CHANGE` calls `onExit()` on the old stack then `onEnter()` on the new state; `PUSH` calls `onPause()` on the state below and `POP` calls `onResume()` on the new top state. When the top state is an overlay, `render()` draws the state below first. These behaviors are real lifecycle semantics, not just inheritance relations on the picture.

2. Ownership boundary of the gameplay states

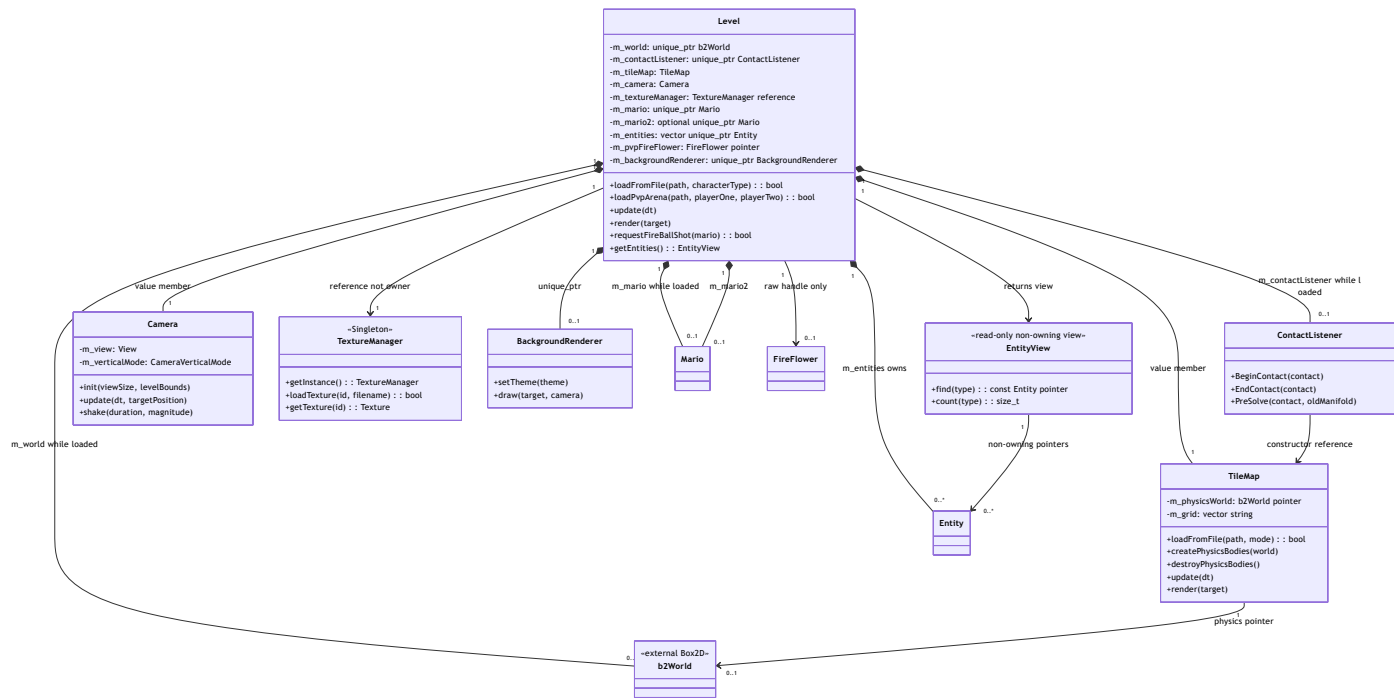
`PlayState` is the only campaign state owning the current level, HUD, two by-value `InputHandler`s (the second handler is for co-op), session progress, and event subscription tokens. `PvpPlayState` has an arena `Level` and two input handlers. Neither state owns `Mario` directly; `Level` is the owner.



In `PlayState::onEnter()` , four subscriptions are kept in `m_eventSubscriptions` ; `onExit()` releases them before stopping the music. On receiving `GAME_PAUSED` , the state enqueues `PauseState` through `GameManager` instead of owning the overlay itself. `HUD` holds a `const Mario&` and optionally a second `const Mario*` , so the arrow to `Mario` means reading data, not ownership.

3. Level, physics, and the entity collection

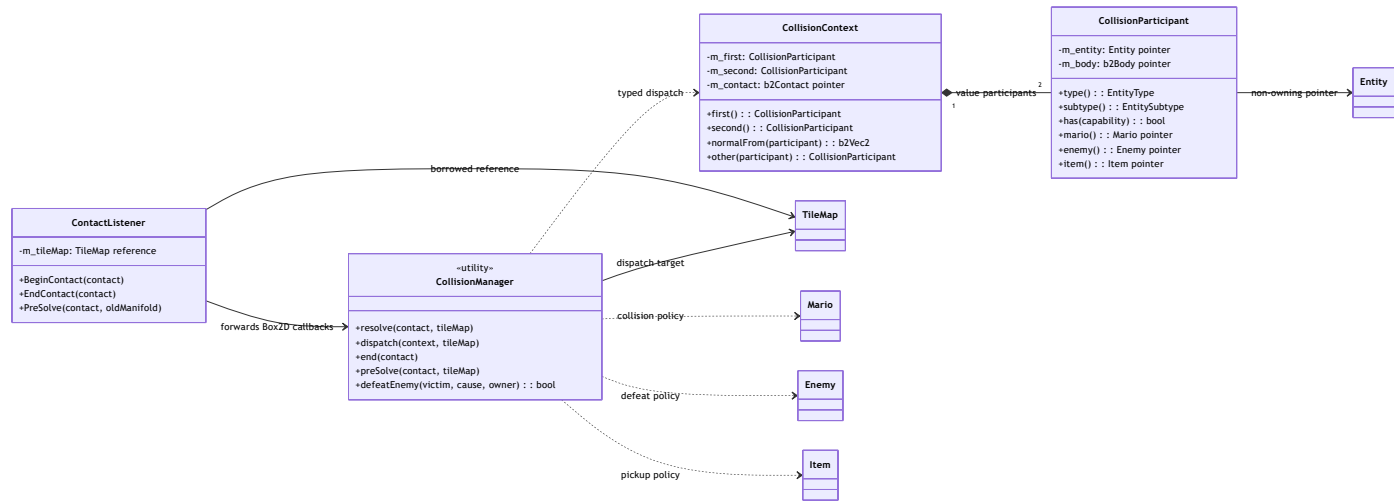
`Level` is the ownership boundary of a stage: the `Box2D` world and listener are `unique_ptr` s, `TileMap` / `Camera` are by-value members, the primary and second `Mario` (co-op/PvP) are `unique_ptr` s, and spawned entities are gathered in a `vector<unique_ptr<Entity>>` . `TextureManager&` is a reference taken from the global `Singleton`; `Level` does not release it. `m_pvpFireFlower` is only a raw handle to a flower already inside `m_entities` .



Level::spawnEntitiesFromTileMap() creates a local EntityFactory , calls create(SpawnRequest, SpawnContext) , attaches the TextureManager , and pushes the result into m_entities ; Mario is created created separately beforehand. Therefore the Level – Entity relation in the picture is true ownership, while getEntities() only returns a view that expires when Level loads/updates/mutates.

4. Collision callback boundary

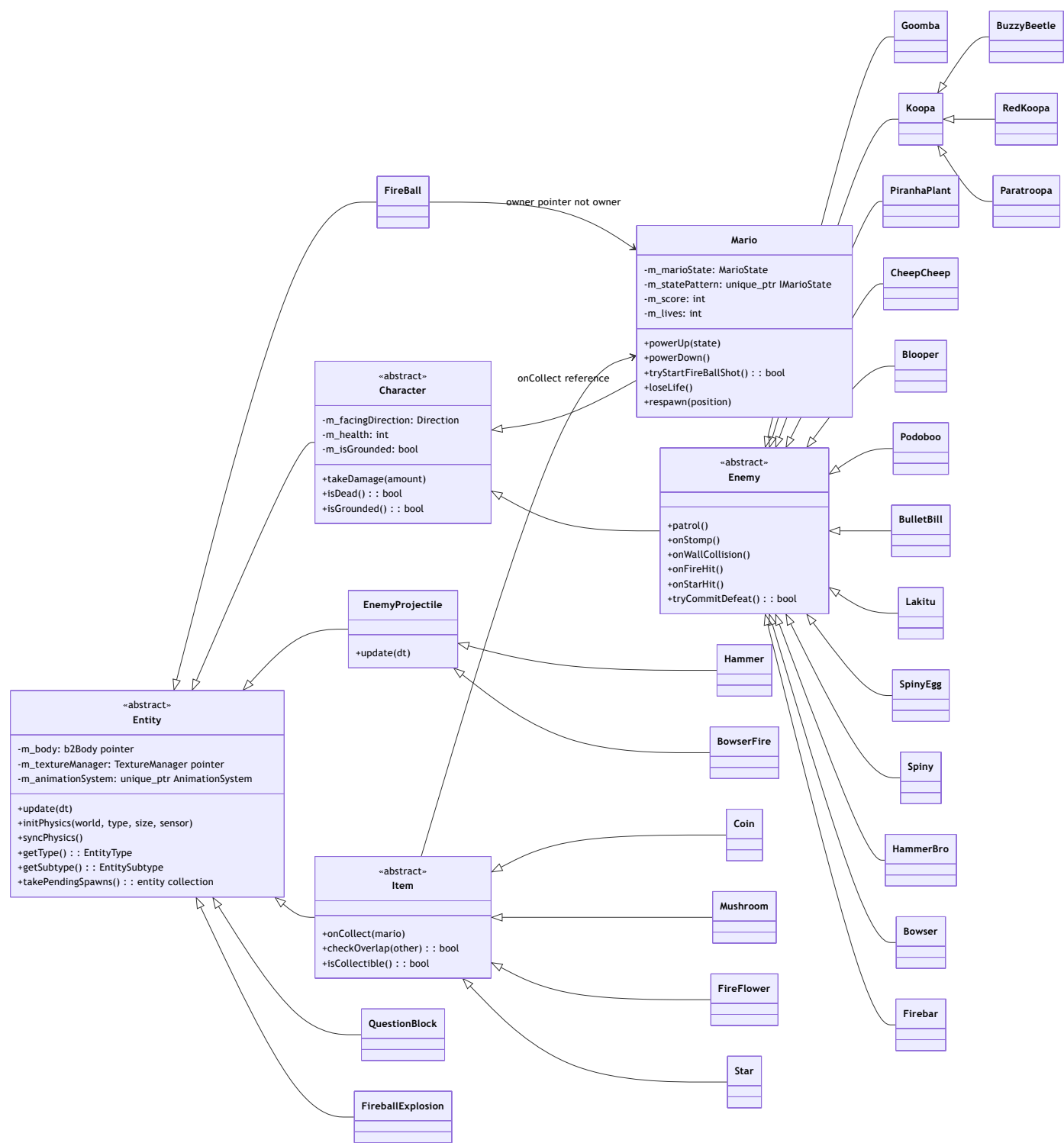
Box2D calls ContactListener ; the listener forwards the callback to CollisionManager . This utility creates/receives a CollisionContext containing two type-checked CollisionParticipant s. The participants merely point at live entities and bodies; they never change Level's ownership.



This is dependency/callback, not a new object graph: ContactListener 's constructor takes a TileMap& ; CollisionManager has a deleted constructor and only provides static helpers. Compatibility overloads like isMario() or isItem() on Entity are not drawn as new subclasses.

5. Entity inheritance tree

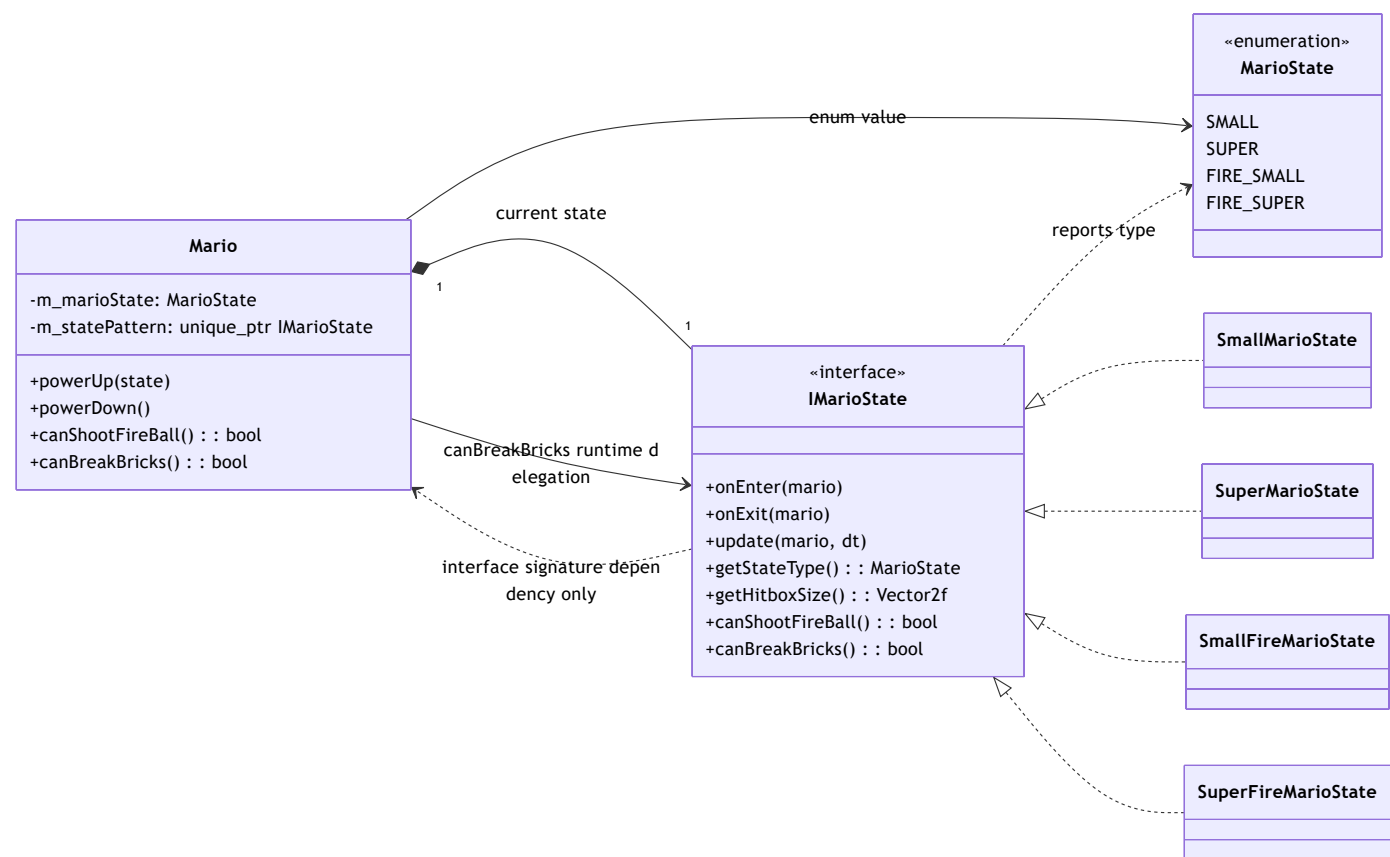
This diagram separates the hierarchy from Level's ownership. The enemy and item branches list the current concrete headers; other direct world objects such as Elevator , Springboard , Toad , ScorePopup , BlockDebris , BowserAxe , and BulletBillLauncher are still Entity s but are omitted from the enemy/item slice.



Entity receives a non-owning TextureManager* ; AnimationSystem is the state held by the entity via unique_ptr . Enemy and Item are polymorphic behavior interfaces (patrol/defeat and onCollect), while runtime collision identity still relies on EntityType , EntitySubtype , and capability bits. Therefore the enum identity must not be interpreted as an extra inheritance layer.

6. Mario power-up state

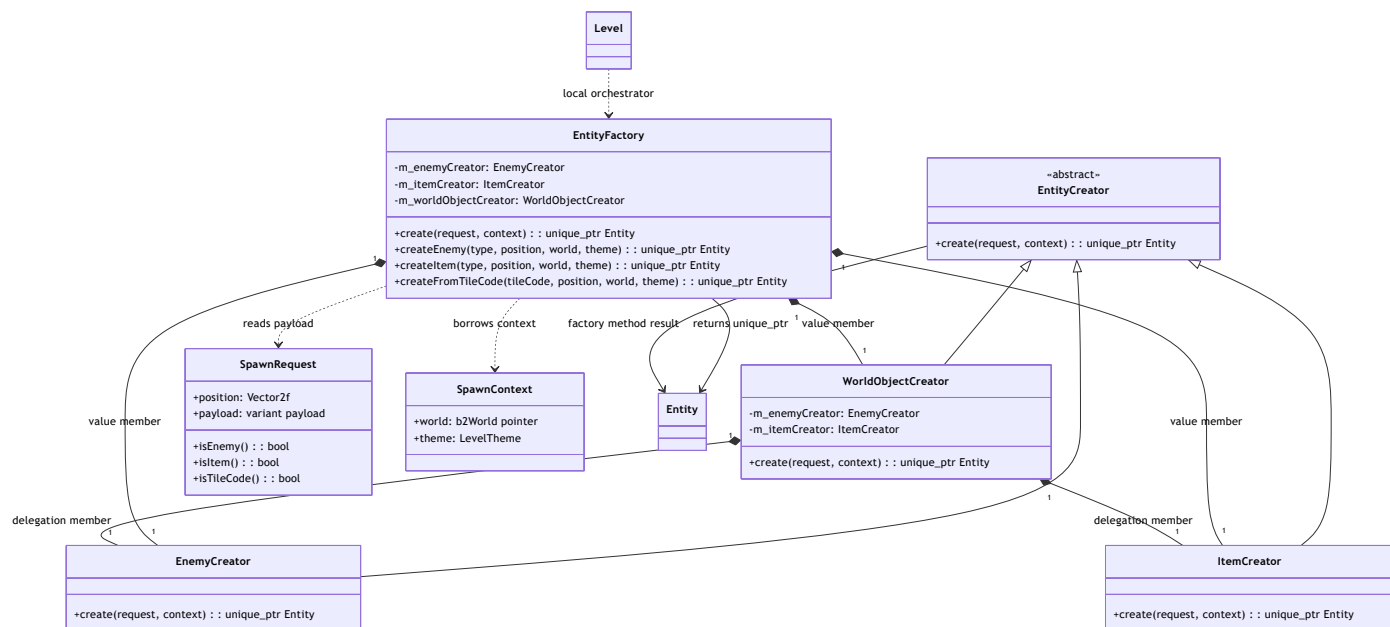
Besides the game's state stack, Mario has its own State Pattern. Mario holds the current state in a std::unique_ptr<IMarioState> and swaps the implementation when applyStateTransition() changes MarioState ; the four concrete states only supply the hitbox, fireball, and brick-breaking rules.



These are two different state seams: `IGameState` controls the lifecycle of game screens; `IMarioState` controls the power-up rules of one Mario. The methods `onEnter`, `onExit`, `update`, `getHitboxSize`, and `canShootFireBall` remain interface signatures, but there is no production call through `m_statePattern` for them in the current `Mario.cpp`. The dashed arrow to `Mario` is therefore only a parameter-type dependency; the single runtime arrow in this slice is `Mario::canBreakBricks()` calling `m_statePattern->canBreakBricks()`. Do not merge the two state seams into one shared hierarchy, and do not assign `SaveManager` to the State Pattern.

7. Factory Method: request -> creator -> entity

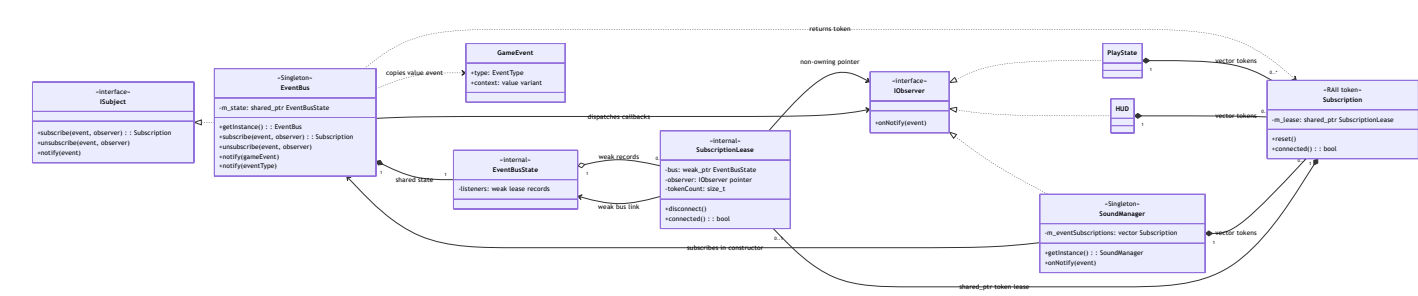
`EntityFactory::create()` is the canonical entry point. A `SpawnRequest` carries exactly one variant payload among `EnemyType`, `ItemType`, or a tile code; `SpawnContext` passes only a non-owning `b2World*` and the theme. `EntityFactory` holds three creators by value; `WorldObjectCreator` in turn holds two stateless creators by value. There is no Singleton label on the factory: the old static helpers merely forward to the default orchestrator.



In `Level::spawnEntitiesFromTileMap()`, the factory creates the object and only then does `Level` attach the texture and tile map for enemies and move the `unique_ptr` into the collection. Only `Level` receives ownership of the result; creators keep no entity after returning it.

8. EventBus and Observer with the RAII token

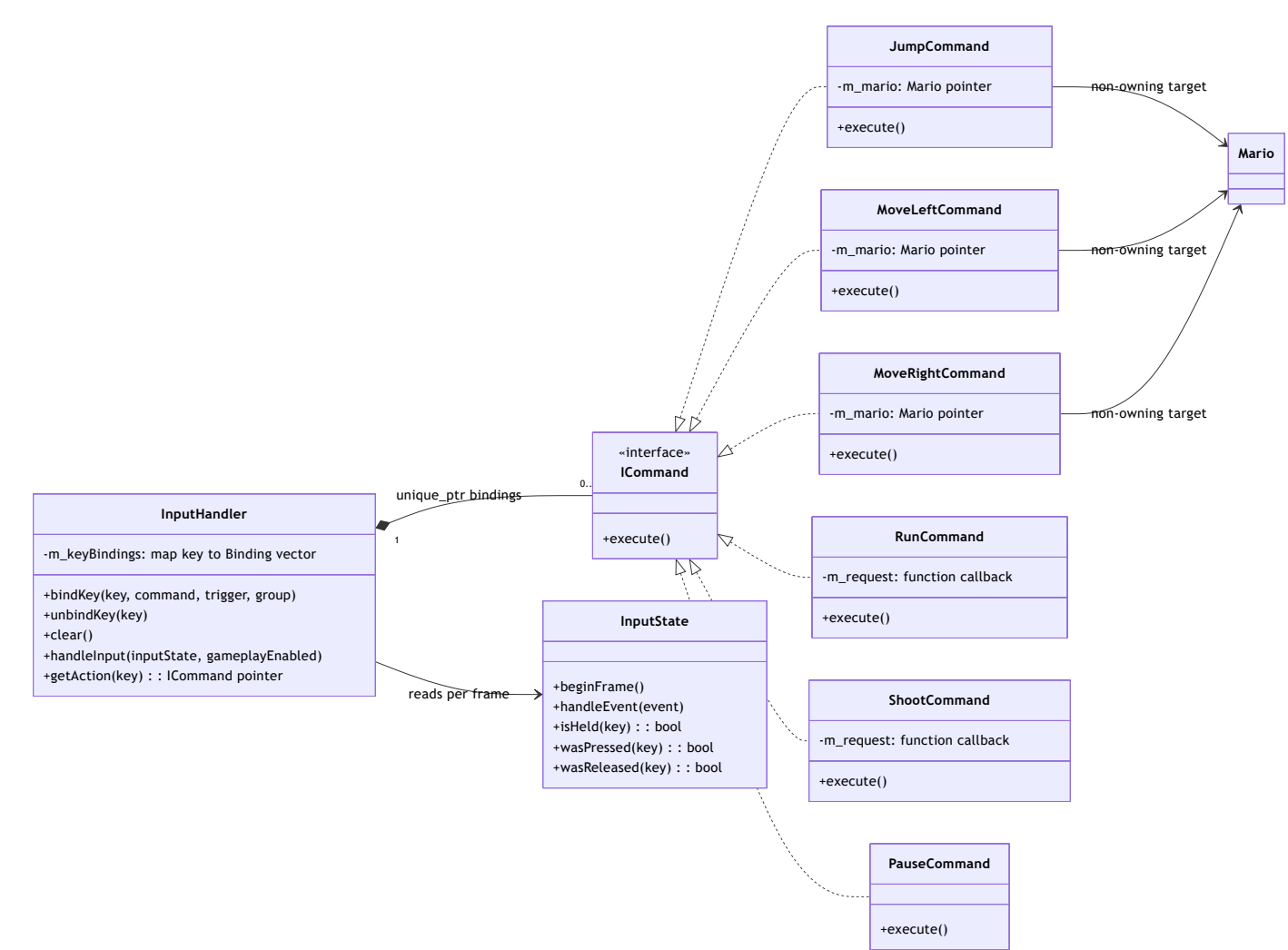
EventBus implements ISubject and is a Singleton. subscribe() returns a move-only Subscription; the token holds a shared_ptr<SubscriptionLease>, while EventBus state stores only weak lease records. The IObserver* inside a lease is a non-owning pointer. When the last token is destroyed or reset(), the registration is disconnected.



EventBus::notify(const GameEvent&) snapshots the listeners and re-checks the lease before each callback, so a callback may reset() another token without corrupting the loop. PlayState::onEnter() registers four events; HUD is also an observer and keeps its own tokens. SoundManager is the third IObserver Singleton: Game obtains the instance at the composition root, the constructor subscribes to 21 events with EventBus and keeps the tokens so onNotify() can play SFX. The EventType-only notify remains a compatibility overload; the canonical payload is the value-only GameEvent.

9. Command and input rebinding

InputHandler owns commands via unique_ptr inside bindings; rebinding the same key/trigger/group replaces the old object. handleInput() reads the InputState, executes Pressed/Released/Held commands, and picks the newest horizontal/vertical command. Concrete commands hold a non-owning Mario pointer or receive a callback; in particular ShootCommand only issues a request and never creates the FireBall itself.



`PlayState::rebindCommands()` builds the concrete commands with
`m_level->getMario()` and maps the shoot callback to
`Level::requestFireBallShot()` . This is a dependency created at the call
site, not field ownership from the command to Level.

10. Source evidence and representation limits

Diagram relation/contract	Current evidence
Game loop delegates to the manager	include/core/Game.h:14 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/Game.h#L14), src/core/Game.cpp:78 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/Game.cpp#L78), src/core/Game.cpp:105 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/Game.cpp#L105), src/core/Game.cpp:129 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/Game.cpp#L129)
Singleton, deferred state stack, value SaveManager	include/core/GameManager.h:23 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/GameManager.h#L23), include/core/GameManager.h:72 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/GameManager.h#L72), include/core/GameManager.h:74 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/GameManager.h#L74), include/core/SaveManager.h:22 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/SaveManager.h#L22), src/core/GameManager.cpp:27 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L27), src/core/GameManager.cpp:76 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L76), src/core/GameManager.cpp:88 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L88)
change/push/pop lifecycle and overlay rendering	src/core/GameManager.cpp:44 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L44), src/core/GameManager.cpp:57 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L57), src/core/GameManager.cpp:64 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L64), src/core/GameManager.cpp:109 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L109)
Concrete game states and the state contract	include/states/IGameState.h:19 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/IGameState.h#L19), include/states/PlayState.h:23 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PlayState.h#L23), include/states/PvpPlayState.h:24 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PvpPlayState.h#L24), include/states/PauseState.h:13 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PauseState.h#L13)

Diagram relation/contract	Current evidence
PlayState OWNS Level/HUD/handlers/tokens	<u>include/states/PlayState.h:77</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PlayState.h#L77), <u>include/states/PlayState.h:78</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PlayState.h#L78), <u>include/states/PlayState.h:82</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PlayState.h#L82), <u>include/states/PlayState.h:106</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PlayState.h#L106), <u>include/ui/HUD.h:37</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/ui/HUD.h#L37), <u>src/states/PlayState.cpp:281</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/states/PlayState.cpp#L281).
Level ownership and non-owning handles	<u>include/level/Level.h:30</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/level/Level.h#L30) <u>include/level/Level.h:210</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/level/Level.h#L210) <u>include/level/Level.h:211</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/level/Level.h#L211) <u>include/level/Level.h:212</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/level/Level.h#L212) <u>include/level/Level.h:215</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/level/Level.h#L215) <u>include/level/Level.h:218</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/level/Level.h#L218) <u>src/level/Level.cpp:174</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L174) <u>src/level/Level.cpp:175</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L175)
Level factory call and entity adoption	<u>src/level/Level.cpp:685</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L685) <u>src/level/Level.cpp:705</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L705) <u>src/level/Level.cpp:730</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L730) <u>src/level/Level.cpp:737</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L737)

Diagram relation/contract	Current evidence
SoundManager Singleton observer and subscription seam	<u>include/core/SoundManager.h:116</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/SoundManager.h#L116) <u>include/core/SoundManager.h:127</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/SoundManager.h#L127) <u>include/core/SoundManager.h:251</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/SoundManager.h#L251) <u>src/core/SoundManager.cpp:40</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/SoundManager.cpp#L40) <u>src/core/SoundManager.cpp:45</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/SoundManager.cpp#L45) <u>src/core/SoundManager.cpp:48</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/SoundManager.cpp#L48) <u>src/core/SoundManager.cpp:105</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/SoundManager.cpp#L105) <u>src/core/SoundManager.cpp:109</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/SoundManager.cpp#L109) <u>src/core/Game.cpp:62</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/Game.cpp#L62), <u>src/core/Game.cpp:65</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/Game.cpp#L65).
Command ownership and dispatch	<u>include/patterns/InputHandler.h:40</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/patterns/InputHandler.h#L40) <u>include/patterns/InputHandler.h:86</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/patterns/InputHandler.h#L86) <u>include/patterns/ICommand.h:16</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/patterns/ICommand.h#L16) <u>src/patterns/InputHandler.cpp:18</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/patterns/InputHandler.cpp#L18) <u>src/patterns/InputHandler.cpp:46</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/patterns/InputHandler.cpp#L46) <u>src/states/PlayState.cpp:62</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/states/PlayState.cpp#L62)

The diagrams do not attempt to list every private field, every tile object, or every gameplay enum. What is omitted is implementation detail that does not change the ownership/contract boundaries drawn; a class not appearing must not be inferred to be non-existent. A Mermaid parser/renderer is not

installed in the workspace, so the document's local verification relies on standard `classDiagram` blocks, class IDs free of dangerous namespace/generic characters, and delimiter balance checks.